Kedar Joshi

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y2jcodebreaker.github.io | scholar.google.com/citations?user=ukrpKLEAAAAJ | orcid.org/0009-0009-9369-7524

EDUCATION

University of Southern California, Los Angeles, CA **Aug 2026 – May 2028**
MS in Computer Science

Manipal University Jaipur, Jaipur, India **Graduated May 2026**
Bachelor of Technology in Computer Science with *Distinction* (CGPA 8.88/10)
Relevant Courses: Data Structure and Algorithms, Artificial Intelligence and Soft Computing, Data Science and Machine Learning

PUBLICATIONS

The Misallocation of Precision in Quantized Language Models **Preprint, 2026**
Kedar Joshi. *Sole author.*

- The activation-covariance Hessian trace used by GPTQ, OmniQuant and ResQ to allocate bit-width is anti-correlated with functional importance across all 64 sub-layers of Llama-3.1-8B ($\rho = -0.497$, $p < 0.001$).
- At an identical 6-of-32 FP16 budget and identical attention/MLP composition, Hessian-selected layers recover 7.7% of ARC quantization damage against 53.8% for probe-selected layers ($p = 0.002$). Localized to named sub-layers against a null measured on the same examples, under BH-FDR across 192 tests.
- The inversion replicates on Mistral-7B and under GPTQ; the Llama-derived layer set transfers to Mistral unchanged. A 6-of-32 mixed-precision model is statistically indistinguishable from FP16 on all three benchmarks at 42% memory.

Contrastive Activation Decomposition: Activation Steering Mitigation by Leveraging Bias Geometry **Preprint, 2026**
Kedar Joshi, Kripabandhu Ghosh. *First author.*

- Diagnostic that prescribes intervention rank and target layers before any steered generation, by decomposing contrastive activation differences.
- Bias geometry varies by type: housing discrimination is near rank-1 ($EVR_1 = 67.9\%$), medicalization bias is diffuse ($EVR_1 = 10.2\%$), explaining why rank-1 steering methods fail on one and succeed on the other.
- Matches CAA's debiasing on AccessEval ($d = 0.337$ vs 0.341) with no per-task tuning, at a fifth of the output-quality cost; validated on Llama, Qwen, Gemma, and Mistral models.

Don't Judge a Book by its Cover: Testing LLMs' Robustness Under Logical Obfuscation **EACL 2026 (main conference)**
Abhilekh Borah, Shubhra Ghosh*, **Kedar Joshi***, Aditya Kumar Guru*, Kripabandhu Ghosh. **Equal contribution.*

- Introduced Logifus, a structure-preserving logical obfuscation algorithm, and LogiQAte, a 1,108-question diagnostic benchmark across four reasoning tasks.
- Frontier models degrade by 28.8% on average under logically equivalent rewrites, up to −86% in the worst case, with a 50–80% collapse in late-layer next-token confidence.
- Code: github.com/abhilekhborah/logical-obfuscation · Data: huggingface.co/datasets/abhilekhborah/LogiQAte

Integrating GLCM Texture Analysis for Improved Deepfake Detection on CelebDF(v2) Dataset **ACOIT 2024, IEEE**
Kedar Joshi, Aditya Sinha. *First author.*

- Texture and frequency features with gradient-boosted trees, detecting manipulation without a CNN forward pass.
- Targets resource-constrained and edge deployment where CNN-based detectors are impractical.
- ieeexplore.ieee.org/document/10939531

PROFESSIONAL EXPERIENCE

Indian Institute of Science Education and Research (IISER), Kolkata, India **May 2025 – Nov 2025**
Research Intern – LLM Evaluation and Robustness

- Evaluated six state-of-the-art models (GPT-4o, GPT-5, o4-mini, Claude, Gemini, Qwen) for reasoning robustness under logical obfuscation; published at EACL 2026 (see Publications).

India Meteorological Department, Bhopal, India **Nov 2024 – Feb 2025**
Research Intern – AI/ML for Atmospheric Sciences

- Filed a patent application for automated physics-informed neural architecture search for single-radar 3D atmospheric velocity field reconstruction.
- 3D wind retrieval has required two or more Doppler radars since the 1960s, because a single radar measures only the radial velocity component. This is the first method to reconstruct the full field from one radar, using conservation laws in place of the geometric constraints the second radar provided.
- Architecture search combining DARTS and Bayesian optimization, treating physics compliance as a third objective alongside accuracy and efficiency, across 245K to 50M+ parameters.

ACADEMIC PROJECTS AND PAPERS

Perturbation-Aware Fine-Tuning (PAFT) for Reducing LLM Memorization

- Training-time data augmentation method that rewrites training examples to reduce memorization, with MemGenTOI, a metric for the memorization-generalization trade-off.

TECHNICAL SKILLS

- **Languages & Tools:** Python, C, PyTorch, TensorFlow, CUDA, Git, Prover9
- **Areas:** LLM evaluation, mechanistic interpretability, quantization, physics-informed neural networks, neural architecture search, benchmark development